

Before we start:

```
$ git clone https://github.com/PentesterLab/codereview-php
```

```
$ git clone https://github.com/PentesterLab/codereview-golang
```

```
https://github.com/snyff/Talks/blob/master/Intro Code Review Owasp BA.pdf
```





Web Security Code Review Workshop

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ABOUT ME:

- Founder and CEO of PentesterLab
- Ex: Pentester, Code Reviewer, AppSec Engineer





- Online Platform to Learn Code Review and Web Hacking / Web Penetration Testing
- Online Live Training Sessions on Web Security Code Review



This WorkShop

- Introduction
- Routing
- Patterns
- CVE Analysis
- CVE-2008-1930
- Conclusion
- Hands-On Code Review



INTRODUCTION

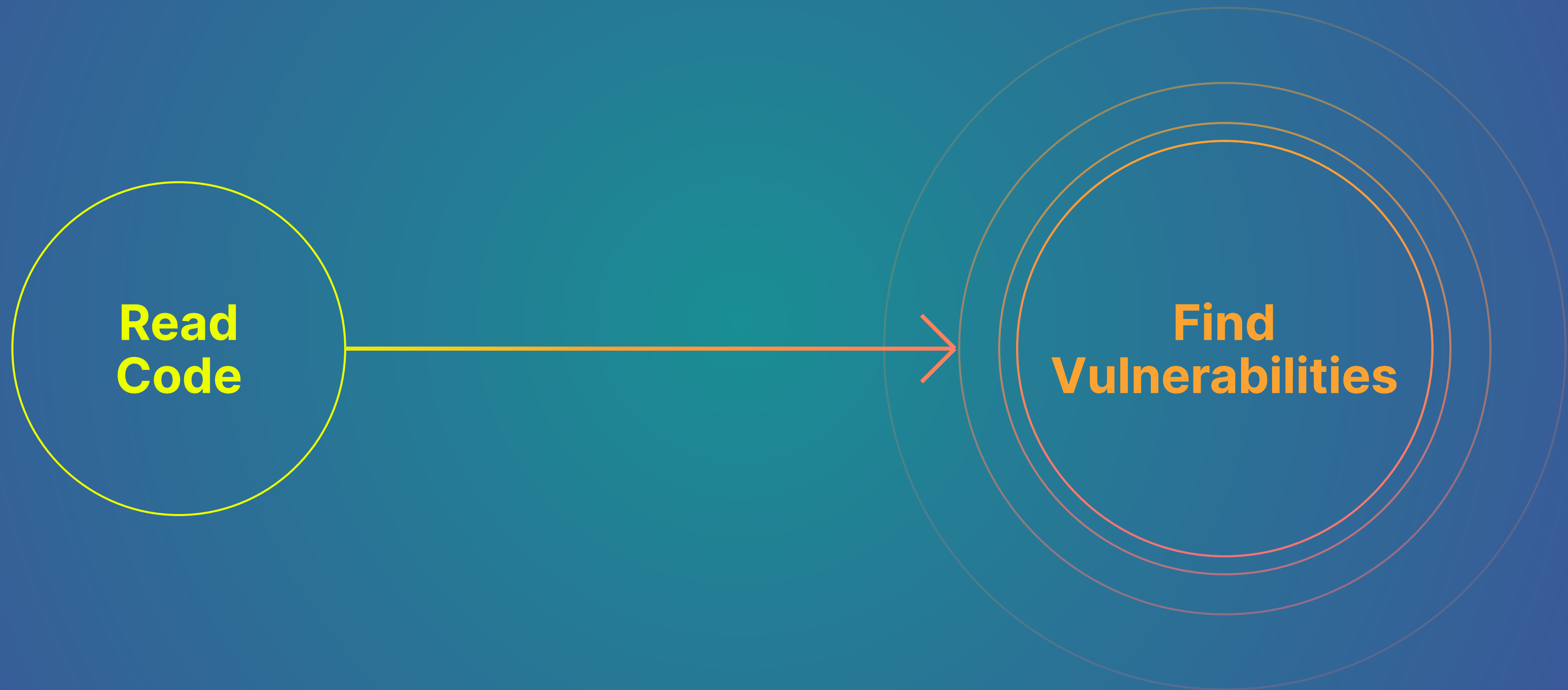


Security Code Review is in demand

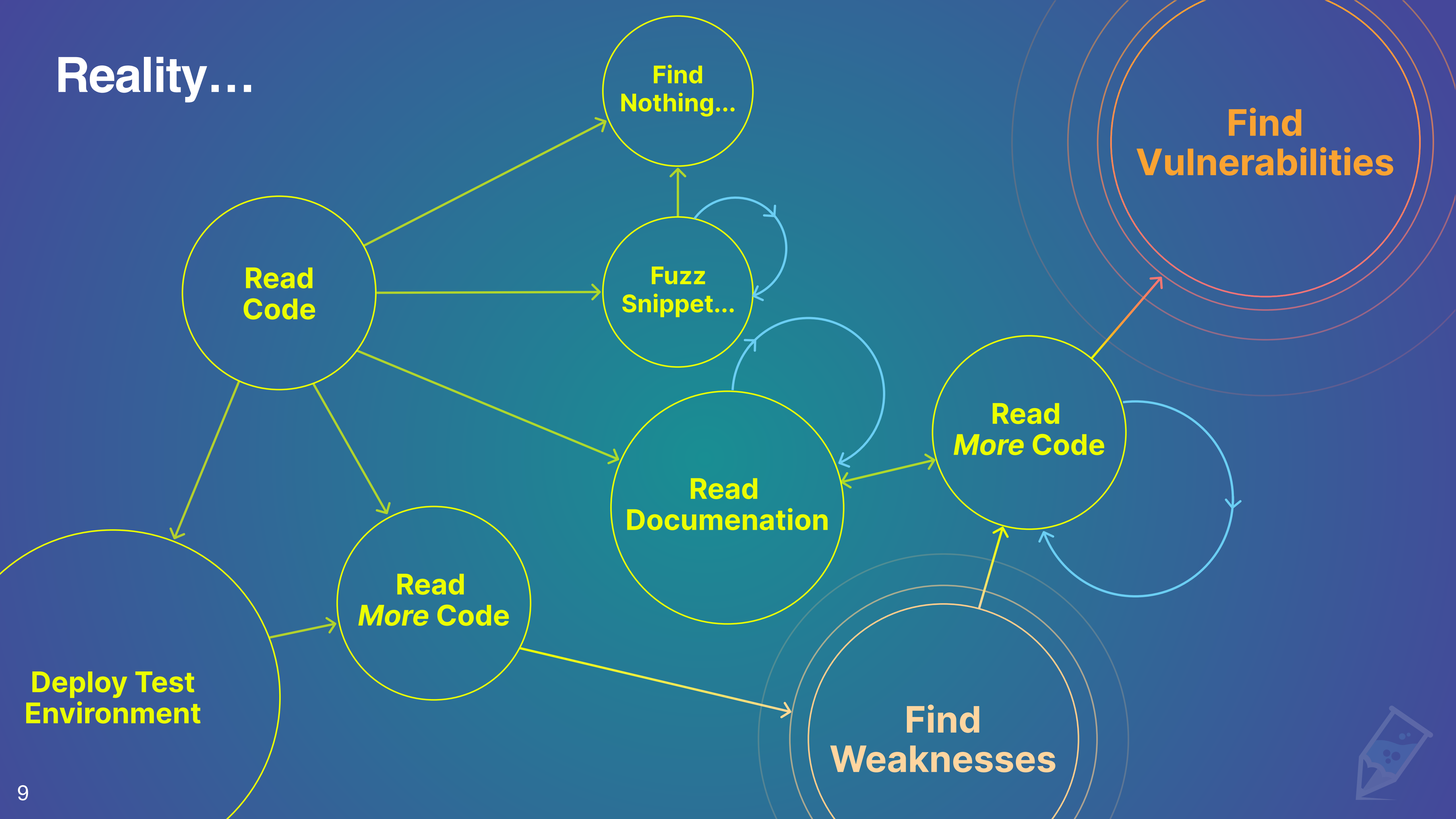
- Ability to find complex bugs
- Ability to find bugs that scanners can't find
- Ability to review changes prior to deployment (Agile, AppSec)
- Ability to find new classes of vulnerabilities
- Powerful skill for:
 - Developers
 - Penetration Testers
 - Security Engineers
 - Vulnerability Researchers / Exploit writers
 - QA/Test Engineer



Expectations...

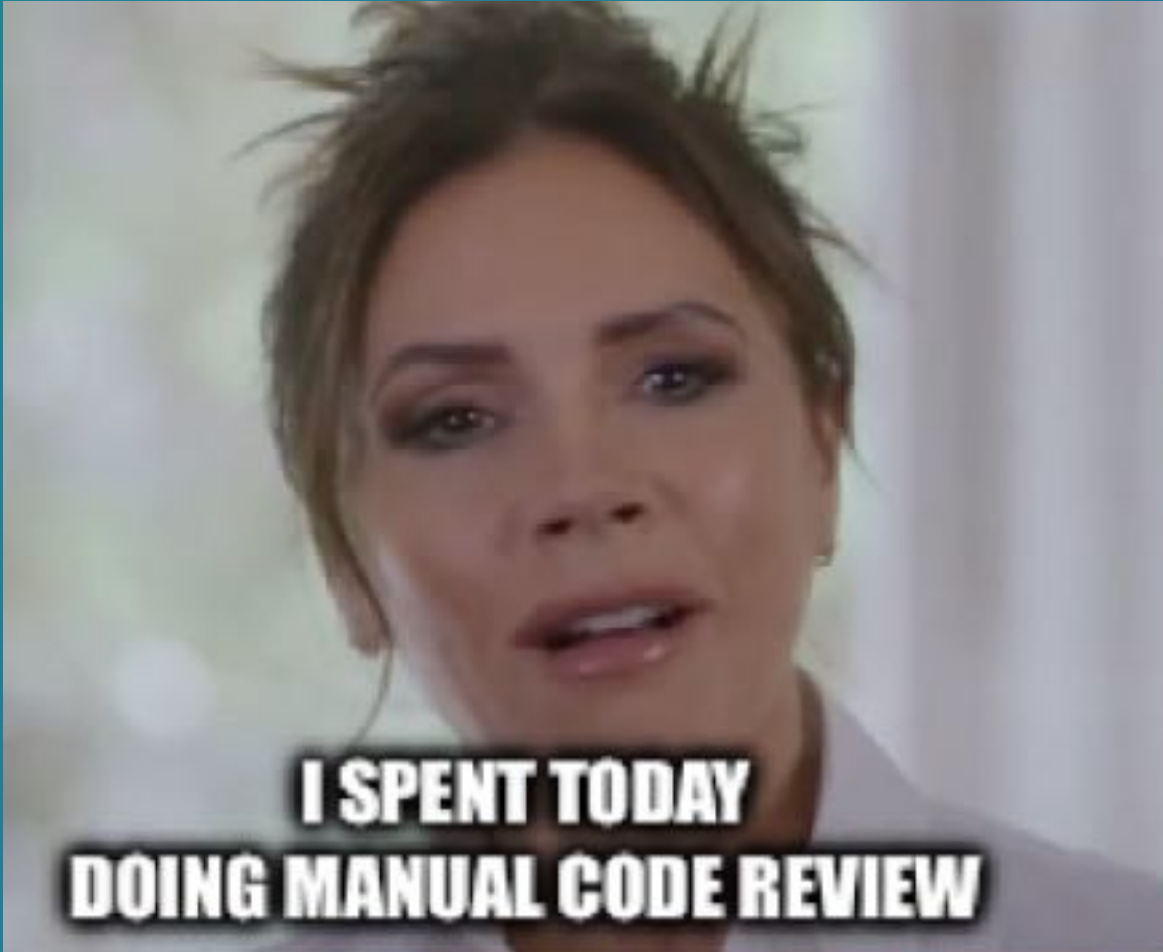


Reality...



Security Code Review...



A woman with dark hair tied back, wearing a white collared shirt, is speaking. The background is a bright, out-of-focus indoor setting.

**I SPENT TODAY
DOING MANUAL CODE REVIEW**

A man with short brown hair is peeking over a white vertical surface, looking towards the left. The background is a plain, light-colored wall.

BE HONEST.

The same woman from the first panel is speaking, looking slightly to her right. The background is the same bright indoor setting.

**I'M
BEING HONEST.**

The same man from the second panel is peeking over the white surface, looking towards the left. The background is the same plain wall.

**WHAT DID
YOU ACTUALLY DO?**

The same woman from the first panel is speaking, looking directly at the camera. The background is the same bright indoor setting.

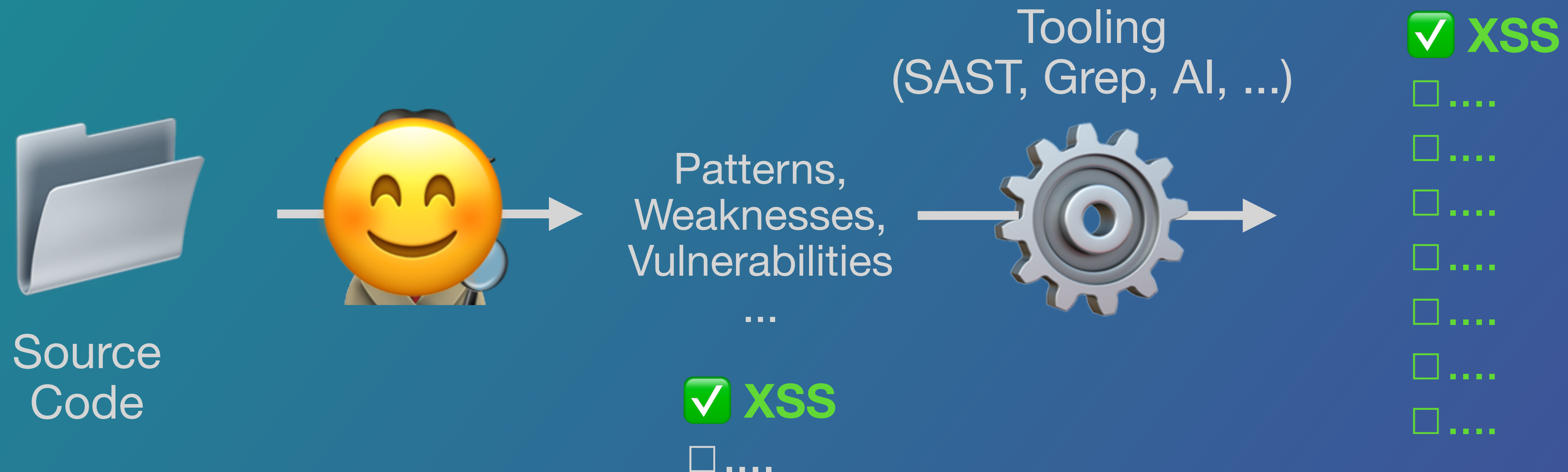
**I RAN GREP A
FEW DOZEN TIMES**

The same man from the second panel is peeking over the white surface, looking towards the left. The background is the same plain wall.

THANK YOU.



Security Code Review...



One of the main advantages of this approach is that it helps identify “unknown unknowns”—issues that automated tools may overlook.



Should you know the language?

A good rule of thumb is that you need to know things that developers don't know:

- Something about a format used?
- A way to bypass a filter?
- Something about threat modelling?
- Something about the language?

The more things you know that the developers don't, the more likely you are to find vulnerabilities



Should you know how to write code?

- It definitely helps!
- You don't need to be a "real" developer but knowing how applications are developed will speed up your work
- The more code you write, the more likely you are to guess:
 - What mistakes developers will make?
 - What shortcuts developers will take?
- The less you know, the more patient you will have to be



Threat modeling

- Key component of code review
- If you don't know what can go wrong, you don't know what to check for
- Knowing about common bug classes for each type of feature or application is key
- Threat modeling gives "direction" to your review

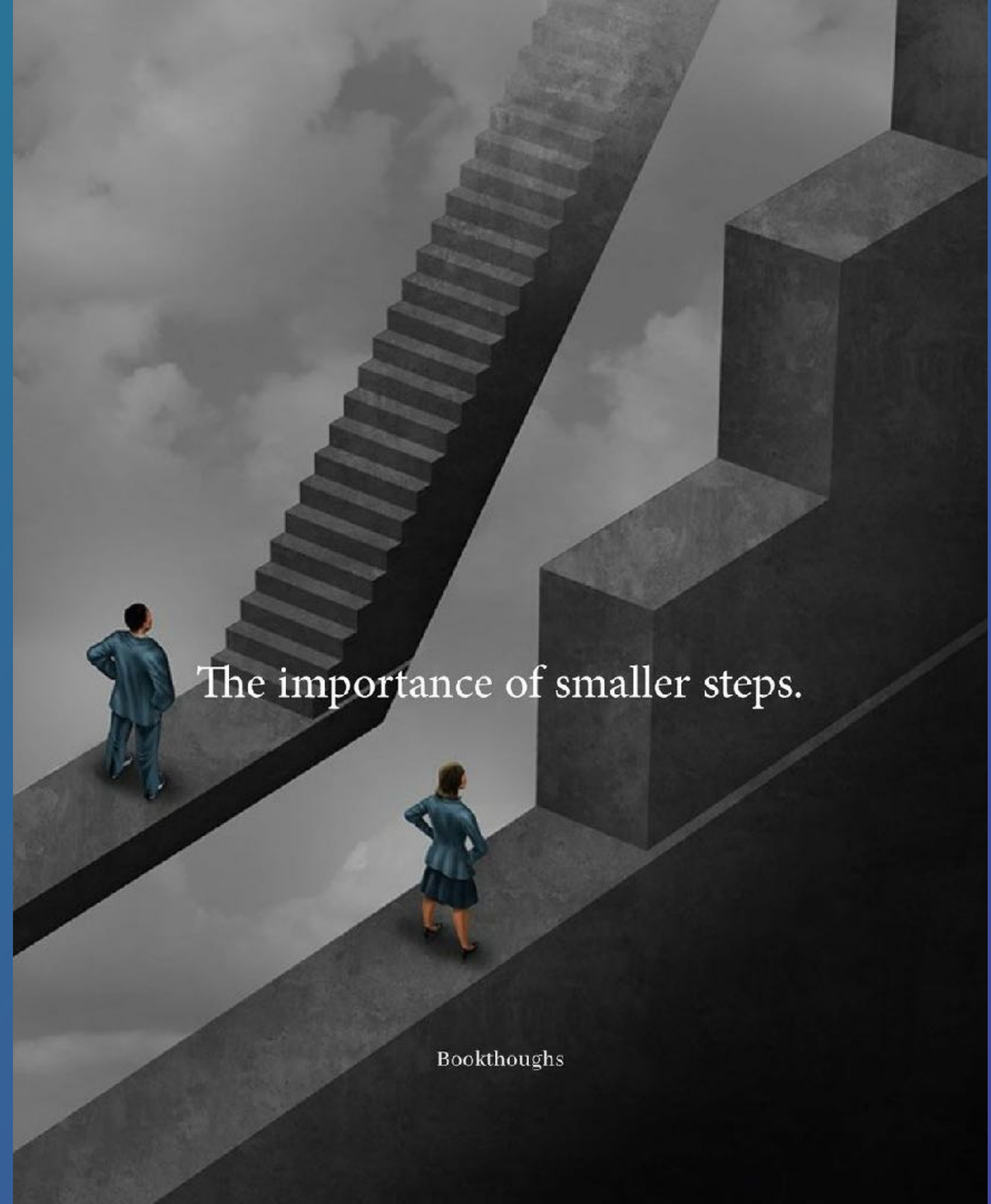


Threat modeling: How to learn?

- There are many methodologies for threat modeling
- For web security code reviews, your best options are to:
 - Read pentest reports
 - Read bug bounty findings and write-ups
 - Follow research presented at conferences
 - Analyze CVE



Picking your targets to learn...



Bookthoughts

Picking your targets to learn...

- You need to find targets that are not too easy
- You need to find targets that are not too hard
- You need to find targets that allow you to grow
- You need to find targets to build resilience



Picking your targets to learn...

1. Snippets
2. Diff from known/public vulnerabilities/CVE
3. Small or simple Libraries
4. Bigger or more complex Libraries
5. Small Applications
6. Larger Applications
7. Hard Targets



Defining Success in Security Code Review

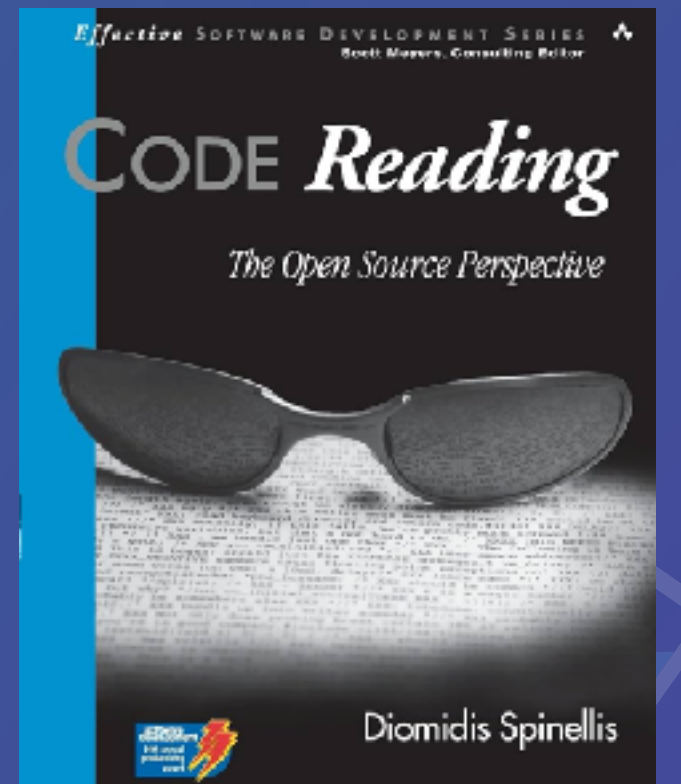
- You don't want to base your success based on the number of vulnerabilities you find or the impact of the vulnerabilities you find (especially when learning)
- Success should be based on:
 - Your progression in understanding a codebase
 - Learning ways a check or filter is implemented
 - Finding small weaknesses or potential improvements
 - Understanding complex patterns
 - Discovering new patterns (with and without security implications)



READING CODE



"An hour of code reading can save you a minute of reading the documentation"



READING THE CODE



Notice:

- Things that are unusual
- When people reinvent the wheel
- Sketchy code
- Complexity
- Unchecked return values
- Checks:
 - What are they trying to prevent?
 - Are they preventing it properly?
 - Is there something else they should take care of but they don't?



Reading code

- Every time you encounter a new function or method:
 - Read the documentation
 - Look for potential security improvements and issues
 - "Fuzz" it (REPL or docker)
 - Keep notes

**Bonus point for doing this over multiple versions of the same method/function...*



Routing



Routing?

How an application maps:

<https://...../foo/1234/bar> to actual code...

- And what is the impact on
 - What you need to review?
 - How you will perform your review?
- Multiple ways to define routing: FS, programmatically, configuration



File System based

- Very common with (old, small, pure, immature) applications (mainly PHP)
- Accessing /index.php is mapped to running the code in the file [WEBROOT]/index.php
- Any file in the web root can potentially be accessed.
- The file's extension or the file's location will decide if the file gets:
 - interpreted/executed: the result of the execution is returned to the client.
 - served: the content of the file is returned to the client.



Programmatically defined

- Code is used to map a route to code

```
package handler

import (
    "net/http"

    "github.com/gin-gonic/gin"
)

func GetRouter() *gin.Engine {
    router := gin.Default()

    router.GET("/", Welcome)
    router.POST("/register", Signup)
    router.POST("/login", Login)

    private := router.Group("/")
    private.Use(Authmiddleware())
    private.GET("/admin/user", Dashboard)
    private.GET("/send", SendMail)
    private.POST("/validate", validate)

    return router
}
```


PATTERNS...



Patterns

When the sage points at the moon, the fool looks at the finger.

- **A lot of issues in security are completely independent of the programming language**
- **In this section, we are going to explore patterns with implementation in multiple programming languages**
- **Make sure you focus on the pattern**



Filter -> Modify -> Use

- The code does three things:
 1. Filters for malicious values
 2. Modifies the value
 3. Uses the value

CAN WE REINTRODUCE SOME
OF THE FILTERED VALUES BACK
USING THE MODIFICATION?



Filter -> Modify -> Use



```
static String validateFileName( String filename )
    throws Exception {
    if( filename == null || filename.trim().isEmpty() ) {
        throw new Exception("Empty File Name");
    }
    final String[] splitpath = filename.split( "[/\\\\\\\\]" );
    filename = splitpath[splitpath.length-1];

    filename = filename.trim();

    // If file name ends with .jsp or .jspx,
    //the user is being naughty!
    if( filename.toLowerCase().endsWith( ".jsp" ) ||
        filename.toLowerCase().endsWith( ".jspx" ) ) {
        throw new Exception("Dangerous extension");
    }

    // Remove any characters that might be a problem.
    return filename.replaceAll("( [?#'\\";] )", "" );
}
```



Filter -> Modify -> Use



```
static String validateFileName( String filename )
    throws Exception {
    if( filename == null || filename.trim().isEmpty() ) {
        throw new Exception("Empty File Name");
    }
    final String[] splitpath = filename.split( "[/\\\\\\]" );
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    if( filename.toLowerCase().endsWith( ".jsp" ) ||
        filename.toLowerCase().endsWith( ".jspx" ) ) {
        throw new Exception("Dangerous extension");
    }

    // Remove any characters that might be a problem.
    return filename.replaceAll("( [?#'\\";] )", "" );
}
```

1



Filter -> Modify -> Use



```
static String validateFileName( String filename )
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    }

    // Remove any characters that might be a problem.
    return filename.replaceAll("( [?#'\\";] )", "" );
}
```

1

2



Filter -> Modify -> Use



```
static String validateFileName( String filename )
    throws Exception {
    if( filename == null || filename.trim().isEmpty() ) {
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        filename.toLowerCase().endsWith( ".jspx" ) ) {
        throw new Exception("Dangerous extension");
    }

    // Remove any characters that might be a problem.
    return filename.replaceAll("( [?#'\\";] )", "" );
}
```

1

2

hack.jsp#



Filter -> Modify -> Use



```
public static String cleanName(String name) {  
    return Normalizer.normalize(  
        HtmlUtil.encode(  
            name.replace(" ", "_")  
                .replace("&", "")  
                .replace("(", "")  
                .replace(")", "")  
                .replace(",", "")  
                .replace("+", "_"), HtmlUtil.ENCODING_TEXT), Normalizer.Form.NFC);  
}
```



Filter -> Modify -> Use



```
public static String cleanName(String name) {  
    return Normalizer.normalize(  
        HtmlUtil.encode(  
            name.replace(" ", "_")  
                .replace("&", "")  
                .replace("(", "")  
                .replace(")", "")  
                .replace(",", "")  
                .replace("+", "_"), HtmlUtil.ENCODING_TEXT), Normalizer.Form.NFC);  
}
```

Filter



Filter -> Modify -> Use



```
public static String cleanName(String name) {  
    return Normalizer.normalize(  
        HtmlUtil.encode(  
            name.replace(" ", "_")  
                .replace("&", "")  
                .replace("(", "")  
                .replace(")", "")  
                .replace(",", "")  
                .replace("+", "_"), HtmlUtil.ENCODING_TEXT), Normalizer.Form.NFC);  
}
```

Modify!



Filter -> Modify -> Use



```
public static String cleanName(String name) {  
    return Normalizer.normalize(  
        HtmlUtil.encode(  
            name.replace(" ", "_")  
                .replace("&", "")  
                .replace("(", "")  
                .replace(")", "")  
                .replace(",", "")  
                .replace("+", "_"), HtmlUtil.ENCODING_TEXT), Normalizer.Form.NFC);  
}
```

The modification may reintroduce things the code filtered...



Filter -> Modify -> Use



```
- return java.text.Normalizer.normalize(  
-     HtmlUtil.encode(  
-         name.replace(" ", "_")  
-         .replace("&", "&")  
-         .replace("(", "(")  
-         .replace(")", ")")  
-         .replace(",", ",")  
-         .replace("+", "+")  
-         .replace(".", "."), HtmlUtil.ENCODING_TEXT), Normalizer.Form.NFC);  
+ return HtmlUtil.encode(Normalizer.normalize(name, Normalizer.Form.NFC)  
+     .replace(" ", "_")  
+     .replace("&", "&")  
+     .replace("(", "(")  
+     .replace(")", ")")  
+     .replace(",", ",")  
+     .replace("+", "+"),  
+     HtmlUtil.ENCODING_TEXT);
```

Normalize then filter/escape



Matching is hard

Ends with, contains, starts with...

- When matching strings without using a Regular Expression, a lot of people get confused on what they are trying to achieve.
- "ends with", "contains", "starts with" may feel like they work similarly for the happy path but they rarely do in reality.



Matching is hard

Ends with, contains, starts with...



```
isLibCurlDomain := strings.Contains(u.UserEmail, "@libcurl.so")
```



Matching is hard

Ends with, contains, starts with...



```
isLibCurlDomain := strings.Contains(u.UserEmail, "@libcurl.so")
```

[louis@libcurl.so.pentesterlab.com](#)



Not matching the correct value...

Ends with, contains, starts with...



```
func authMiddleware(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        tokenString := r.Header.Get("Authorization")

        claims := &Claims{}

        token, err := jwt.ParseWithClaims(tokenString, claims, func(token *jwt.Token) (interface{}, error) {
            return jwtKey, nil
        })

        if err != nil {
            http.Error(w, "Invalid token", http.StatusUnauthorized)
            return
        }

        if !token.Valid {
            http.Error(w, "Invalid token", http.StatusUnauthorized)
            return
        }

        if !strings.Contains(r.URL.String(), "health") && claims.Username != "admin" {
            http.Error(w, "You don't have access to the key", http.StatusUnauthorized)
            return
        }

        next.ServeHTTP(w, r)
    })
}
```



Not matching the correct value...

Ends with, contains, starts with...



```
func authMiddleware(next http.Handler) http.Handler {
    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
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        if !token.Valid {
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        if !strings.Contains(r.URL.String(), "health") && claims.Username != "admin" {
            http.Error(w, "You don't have access to the key", http.StatusUnauthorized)
            return
        }
        next.ServeHTTP(w, r)
    })
}
```



Not matching the correct value...

Ends with, contains, starts with...



```
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        if !strings.Contains(r.URL.String(), "health") && claims.Username != "admin" {
            http.Error(w, "You don't have access to the key", http.StatusUnauthorized)
            return
        }
        next.ServeHTTP(w, r)
    })
}
```

r.URL.String() vs r.URL.Path



Reinventing the wheel!

- Never a good idea (always a bad idea when dealing with crypto)
- For most common operations, programming languages provide built-in functions or methods, such as:
 - String manipulations: uppercase, lowercase, split, cut
 - File manipulations: getting the file extension from a filename, extracting the filename from a path, etc.
- When developers write their own versions, they're likely to overlook odd edge cases that the built-in functions already handle
- Compare the built-in source code with the code written by the developers!



Play Session Injection



```
void save() {  
    [...]  
    try {  
        StringBuilder session = new StringBuilder();  
        for (String key : data.keySet()) {  
            session.append("\u0000");  
            session.append(key);  
            session.append(":");  
            session.append(data.get(key));  
            session.append("\u0000");  
        }  
        String sessionData =  
            URLEncoder.encode(session.toString(), "utf-8");  
        String sign = Crypto.sign(sessionData,  
                                   Play.secretKey.getBytes());  
    }  
}
```



Play Session Injection



```
void save() {  
    [...]  
    try {  
        StringBuilder session = new StringBuilder();  
        for (String key : data.keySet()) {  
            session.append("\u0000");  
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        }  
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            URLEncoder.encode(session.toString(), "utf-8");  
        String sign = Crypto.sign(sessionData,  
                                   Play.secretKey.getBytes());  
    }  
}
```



**They are
reinventing a
serialiser**



Play Session Injection



```
void save() {  
    [...]  
    try {  
        StringBuilder session = new StringBuilder();  
        for (String key : data.keySet()) {  
            session.append("\u0000");  
            session.append(key);  
            session.append(":");  
            session.append(data.get(key));  
            session.append("\u0000");  
        }  
        String sessionData =
```

Session: {"key1": "value1", "key2": "value2"}

becomes "\x00key1:value1\x00\x00key2:value2\x00"



Play Session Injection



```
void save() {  
    [...]  
    try {  
        StringBuilder session = new StringBuilder();  
        for (String key : data.keySet()) {  
            session.append("\u0000");  
            session.append(key);  
            session.append(":");  
            session.append(data.get(key));  
            session.append("\u0000");  
        }  
        String sessionData =
```

Session: {"username": "louis", "email": "louis@pentesterlab.com"}

becomes "\x00username:louis\x00\x00email:louis@pentesterlab.com\x00"



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\\u0000([^\:]*):([^\u0000]*)\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if(firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
                                   Play.secretKey.getBytes())) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```



Play Session Injection

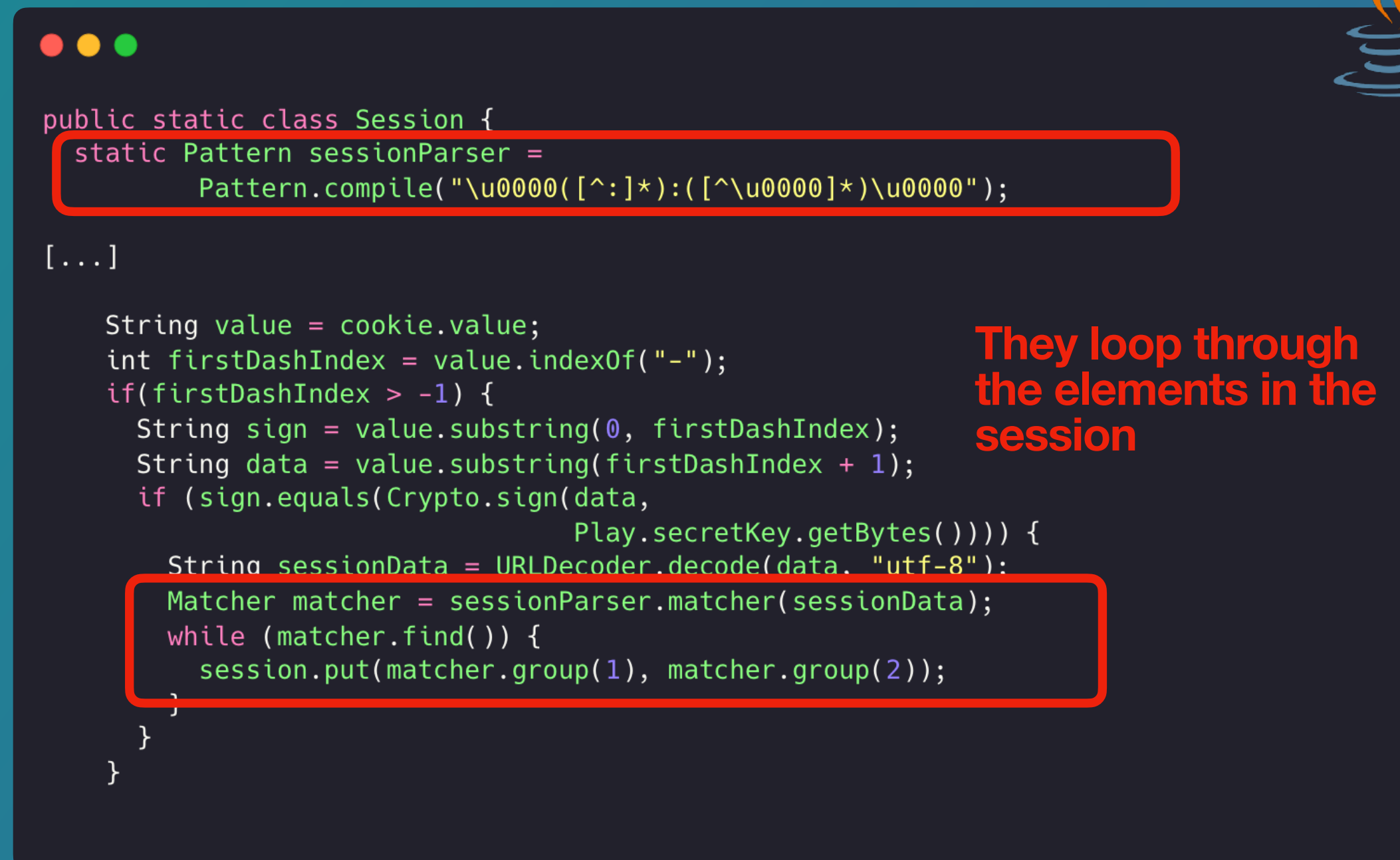


```
public static class Session {  
    static Pattern sessionParser =  
        Pattern.compile("\\u0000([^:]*):([^\u0000]*)\u0000");  
  
    [...]  
  
    String value = cookie.value;  
    int firstDashIndex = value.indexOf("-");  
    if(firstDashIndex > -1) {  
        String sign = value.substring(0, firstDashIndex);  
        String data = value.substring(firstDashIndex + 1);  
        if (sign.equals(Crypto.sign(data,  
                                   Play.secretKey.getBytes())) {  
            String sessionData = URLDecoder.decode(data, "utf-8");  
            Matcher matcher = sessionParser.matcher(sessionData);  
            while (matcher.find()) {  
                session.put(matcher.group(1), matcher.group(2));  
            }  
        }  
    }  
}
```

**They loop
through the
elements in the
session**



Play Session Injection



```
public static class Session {  
    static Pattern sessionParser =  
        Pattern.compile("\\u0000(?:[^\u0000]*)\\u0000");  
    [...]  
  
    String value = cookie.value;  
    int firstDashIndex = value.indexOf("-");  
    if (firstDashIndex > -1) {  
        String sign = value.substring(0, firstDashIndex);  
        String data = value.substring(firstDashIndex + 1);  
        if (sign.equals(Crypto.sign(data,  
            Play.secretKey.getBytes())) {  
            String sessionData = URLDecoder.decode(data, "utf-8");  
            Matcher matcher = sessionParser.matcher(sessionData);  
            while (matcher.find()) {  
                session.put(matcher.group(1), matcher.group(2));  
            }  
        }  
    }  
}
```

They loop through the elements in the session

"\x00key1:value1\x00\x00key2:value2\x00" becomes:

"\x00key1:value1\x00" => session.put("key1" , "value1")

"\x00key2:value2\x00" => session.put("key2" , "value2")



Play Session Injection



```
public void put(String key, String value) {  
    if (key.contains(":")) {  
        throw new IllegalArgumentException(  
            "Character ':' is invalid in a session key.");  
    }  
    [...]  
    if (value == null) {  
        data.remove(key);  
    } else {  
        data.put(key, value);  
    }  
}
```



Play Session Injection



```
public void put(String key, String value) {  
    if (key.contains(":")) {  
        throw new IllegalArgumentException(  
            "Character ':' is invalid in a session key.");  
    }  
    [...]  
    if (value == null) {  
        data.remove(key);  
    } else {  
        data.put(key, value);  
    }  
}
```



**No checks to
prevent separators
(':' or NULL BYTE) in
the value**



Play Session Injection



```
public void put(String key, String value) {  
    if (key.contains(":")) {  
        throw new IllegalArgumentException(  
            "Character ':' is invalid in a session key.");  
    }  
    [...]   
    if (value == null) {  
        data.remove(key);  
    } else {  
        data.put(key, value);  
    }  
}
```

As a client, we most likely only have access to the value.



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\\u0000([^:]*):(^[\\u0000]*)\\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if(firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
                                   Play.secretKey.getBytes())) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

username=[USER-CONTROLLED]



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\\u0000([^:]*):(^[\\u0000]*)\\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if(firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
                                   Play.secretKey.getBytes())) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

username=louis



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
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    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if(firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
                                   Play.secretKey.getBytes())) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

username=louis => session.put("username", "louis")



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\\u0000([^\:]*):([^\u0000]*)\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if(firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
                                   Play.secretKey.getBytes())) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

**username=louis => session.put("username", "louis")
=> "\\x00username:louis\x00"**



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\\u0000([^:]*):(^[\\u0000]*)\\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if(firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
                                   Play.secretKey.getBytes())) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

username=louis\x00\x00username:admin



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\u0000(?:[^\u0000]*)\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if (firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
            Play.secretKey.getBytes()))) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

username=louis\x00\x00username:admin

=> session.put("username", "louis\x00\x00username:admin")



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\u0000(?:[^\u0000]*)\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if (firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
            Play.secretKey.getBytes()))) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

username=louis\x00\x00username:admin

=> session.put("username", "louis\x00\x00username:admin")

=> \x00username:louis\x00\x00username:admin\x00



Play Session Injection



```
public static class Session {
    static Pattern sessionParser =
        Pattern.compile("\u0000(?:[^\u0000]*)\u0000");

    [...]

    String value = cookie.value;
    int firstDashIndex = value.indexOf("-");
    if (firstDashIndex > -1) {
        String sign = value.substring(0, firstDashIndex);
        String data = value.substring(firstDashIndex + 1);
        if (sign.equals(Crypto.sign(data,
            Play.secretKey.getBytes()))) {
            String sessionData = URLDecoder.decode(data, "utf-8");
            Matcher matcher = sessionParser.matcher(sessionData);
            while (matcher.find()) {
                session.put(matcher.group(1), matcher.group(2));
            }
        }
    }
}
```

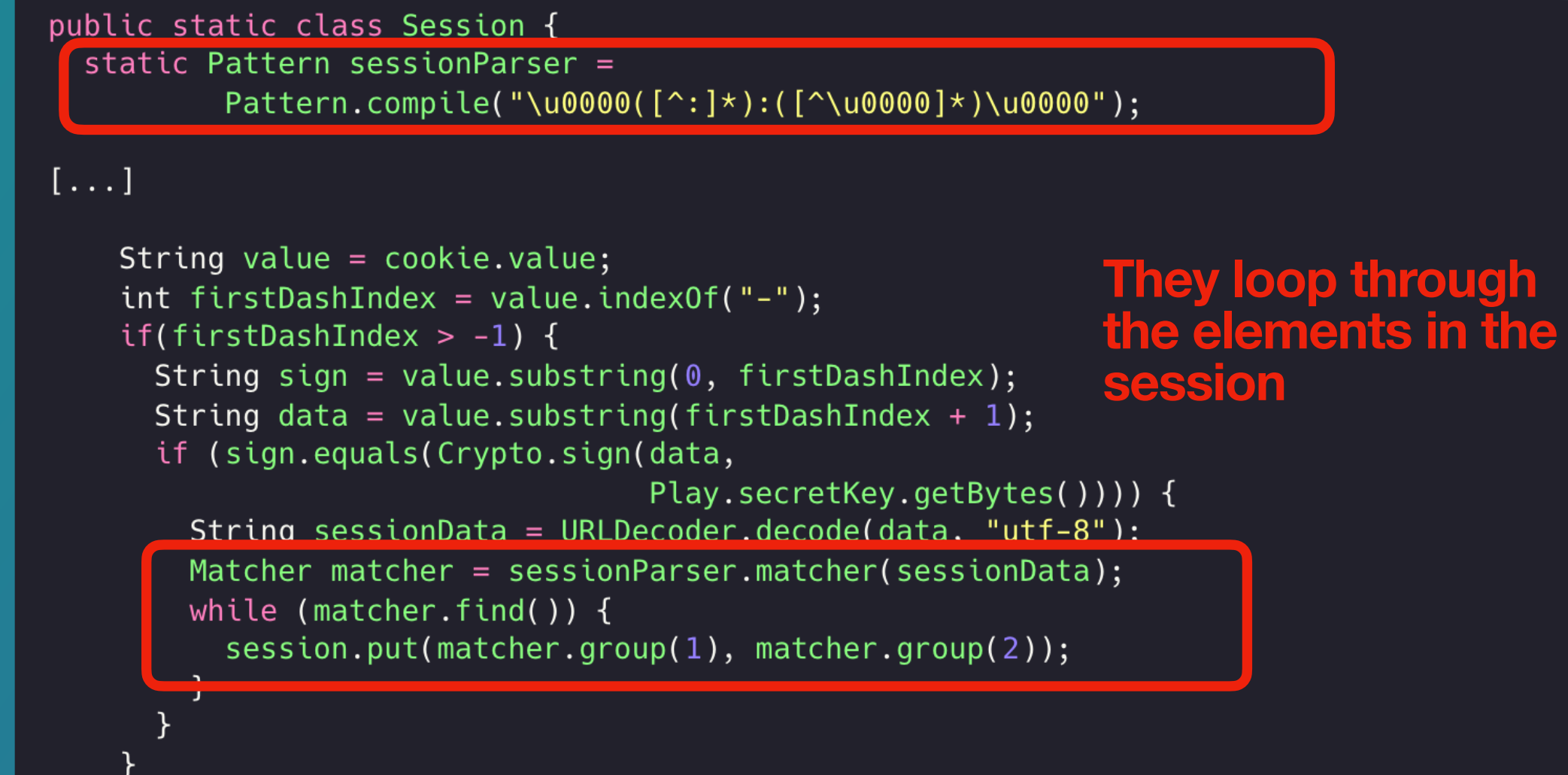
username=louis\x00\x00username:admin

=> session.put("username", "louis\x00\x00username:admin")

=> \x00username:louis\x00\x00username:admin\x00



Play Session Injection



```
public static class Session {  
    static Pattern sessionParser =  
        Pattern.compile("\\u0000(?:[^\u0000]*)\\u0000");  
    [...]  
  
    String value = cookie.value;  
    int firstDashIndex = value.indexOf("-");  
    if (firstDashIndex > -1) {  
        String sign = value.substring(0, firstDashIndex);  
        String data = value.substring(firstDashIndex + 1);  
        if (sign.equals(Crypto.sign(data,  
            Play.secretKey.getBytes())) {  
            String sessionData = URLDecoder.decode(data, "utf-8");  
            Matcher matcher = sessionParser.matcher(sessionData);  
            while (matcher.find()) {  
                session.put(matcher.group(1), matcher.group(2));  
            }  
        }  
    }  
}
```

They loop through the elements in the session

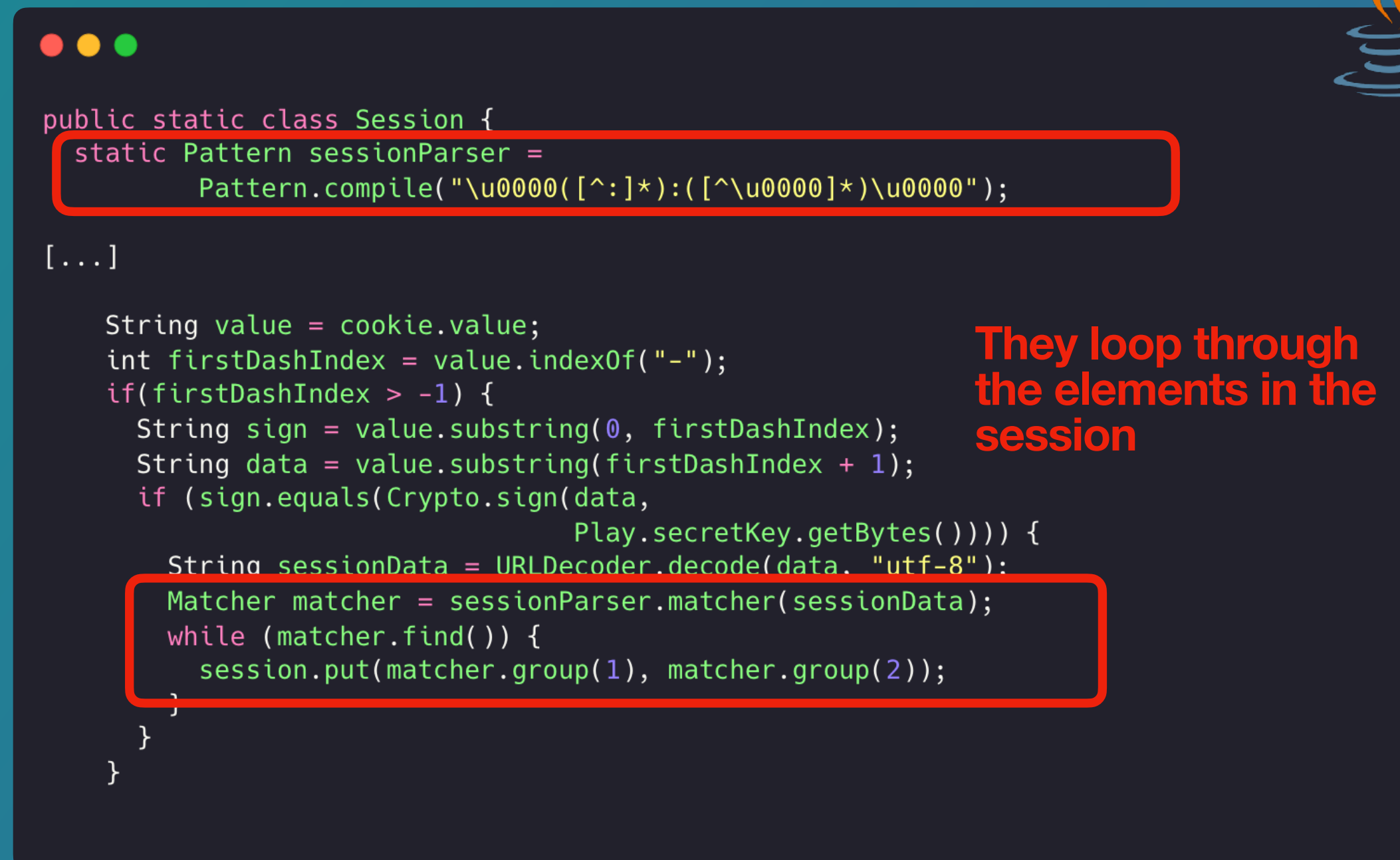
`\x00username:louis\x00\x00username:admin\x00` becomes

`"\x00username:louis\x00"` => `session.put("username" , "louis")`

`"\x00username:admin\x00"` => `session.put("username" , "admin")`



Play Session Injection



```
public static class Session {  
    static Pattern sessionParser =  
        Pattern.compile("\\u0000(?:[^\u0000]*):(?:[^\u0000]*)\\u0000");  
    [...]  
  
    String value = cookie.value;  
    int firstDashIndex = value.indexOf("-");  
    if (firstDashIndex > -1) {  
        String sign = value.substring(0, firstDashIndex);  
        String data = value.substring(firstDashIndex + 1);  
        if (sign.equals(Crypto.sign(data,  
            Play.secretKey.getBytes())) {  
            String sessionData = URLDecoder.decode(data, "utf-8");  
            Matcher matcher = sessionParser.matcher(sessionData);  
            while (matcher.find()) {  
                session.put(matcher.group(1), matcher.group(2));  
            }  
        }  
    }  
}
```

They loop through the elements in the session

`\x00username:louis\x00\x00username:admin\x00` becomes

`"\x00username:louis\x00"` => `session.put("username" , "louis")`

`"\x00username:admin\x00"` => `session.put("username" , "admin")` (OVERWRITE)

CVE ANALYSIS



Why?

- Deliberate practice
- Learning new patterns
- Learning how to fix issues
- Find incomplete patches



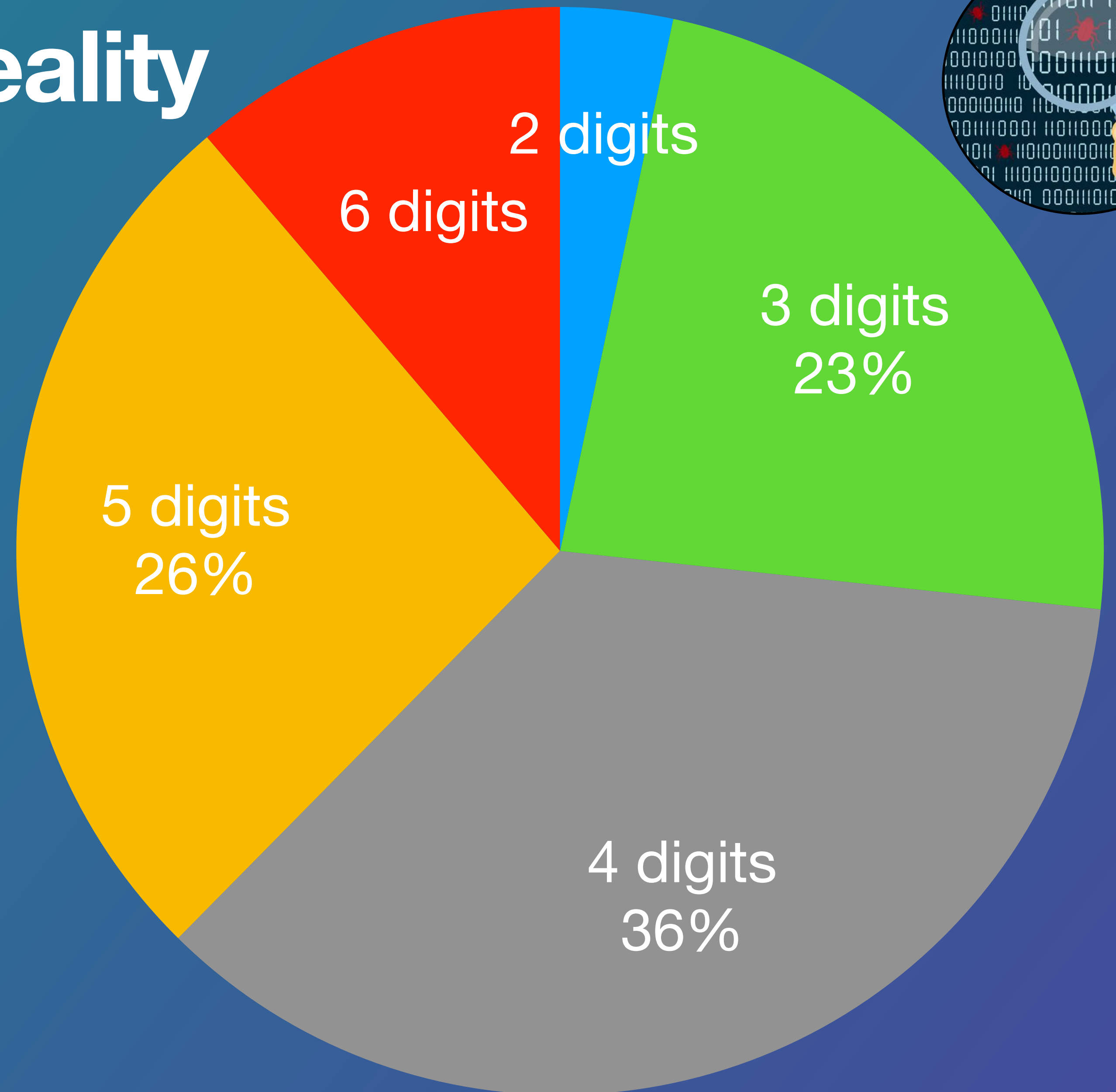
Analysing CVE (Code Review)

- Read the advisory!
- Clone the repository
- Find the tag for the vulnerable and fixed versions
- Extract a patch/diff
- Analyse the patch/diff:
 - What does the vulnerable code look like?
 - What does the fix look like?
 - Is this properly fixed?
 - More vulnerabilities in the same area?



Expectations vs Reality

575 CVEs (diff based on
vulnerable version versus
patched version):



Expectations vs Reality



[VULNERABLE]   [PATCHED]

[VULNERABLE]    [PATCHED]

[VULNERABLE]     [PATCHED]

[VULNERABLE]      [PATCHED]

[VULNERABLE]       [PATCHED]



Methodology



Analysing CVE (Code Review)

<https://github.com/zeromicro/go-zero/security/advisories/GHSA-fgxv-gw55-r5fq>

Authorization Bypass Through User-Controlled Key in go-zero

Critical kevwan published GHSA-fgxv-gw55-r5fq 2 weeks ago

Package

 github.com/zeromicro/go-zero (Go)

Affected versions

< v1.4.4

Patched versions

None

Severity

Critical 9.1 / 10



Analysing CVE (Code Review)

<https://github.com/zeromicro/go-zero/security/advisories/GHSA-fgxv-gw55-r5fq>



```
$ git clone https://github.com/zeromicro/go-zero/security/
```



Analysing CVE (Code Review)

<https://github.com/zeromicro/go-zero/security/advisories/GHSA-fgxv-gw55-r5fq>



```
$ git clone https://github.com/zeromicro/go-zero/
```

```
$ cd go-zero
```

```
$ git tag
```



Analysing CVE (Code Review)

<https://github.com/zeromicro/go-zero/security/advisories/GHSA-fgxv-gw55-r5fq>

Authorization Bypass Through User-Controlled Key in go-zero

Critical kevwan published GHSA-fgxv-gw55-r5fq 2 weeks ago

Package

 github.com/zeromicro/go-zero (Go)

Affected versions

< v1.4.4

Patched versions

None

Severity

Critical 9.1 / 10

```
$ cd go-zero
```

```
$ git tag
```

```
$ git diff v1.4.3...v1.4.4
```




```
$ git clone https://github.com/zeromicro/go-zero/
```

```
$ cd go-zero
```

```
$ git tag
```

```
$ git diff v1.4.3...v1.4.4
```

```
$ git diff v1.4.3...v1.4.4 | grep -i cors
```

```
$ git tag
```

```
$ git diff v1.4.3...v1.5.1 | grep -i cors
```



Analysing CVE (Code Review)

<https://github.com/zeromicro/go-zero/security/advisories/GHSA-fgxv-gw55-r5fq>



```
$ git diff v1.4.3...v1.5.1 | grep -i cors
@@ -535,3 +535,91 @@ func TestServer_WithCors(t *testing.T) {
snyff@snyffs-Air go-zero % git diff v1.4.3...v1.5.1 | grep cors
diff --git a/rest/internal/cors/handlers.go b/rest/internal/cors/handlers.go
--- a/rest/internal/cors/handlers.go
+++ b/rest/internal/cors/handlers.go
diff --git a/rest/internal/cors/handlers_test.go b/rest/internal/cors/handlers_test.go
--- a/rest/internal/cors/handlers_test.go
+++ b/rest/internal/cors/handlers_test.go
    "github.com/zeromicro/go-zero/rest/internal/cors"
```



```
% git diff v1.4.3...v1.5.1 rest/internal/cors/handlers.go
diff --git a/rest/internal/cors/handlers.go b/rest/internal/cors/handlers.go
index e2a64b74..133b47dd 100644
--- a/rest/internal/cors/handlers.go
+++ b/rest/internal/cors/handlers.go
@@ -77,12 +77,19 @@ func checkAndSetHeaders(w http.ResponseWriter, r *http.Request,
    origins []string
    }

    func isOriginAllowed(allows []string, origin string) bool {
-       for _, o := range allows {
-           if o == allOrigins {
+       origin = strings.ToLower(origin)
+
+       for _, allow := range allows {
+           if allow == allOrigins {
+               return true
+           }
+
+           allow = strings.ToLower(allow)
+           if origin == allow {
+               return true
+           }

-           if strings.HasSuffix(origin, o) {
+           if strings.HasSuffix(origin, "."+allow) {
+               return true
+           }
    }
```




```

% git diff v1.4.3...v1.5.1 rest/internal/cors/handlers.go
diff --git a/rest/internal/cors/handlers.go b/rest/internal/cors/handlers.go
index e2a64b74..133b47dd 100644
--- a/rest/internal/cors/handlers.go
+++ b/rest/internal/cors/handlers.go
@@ -77,12 +77,19 @@ func checkAndSetHeaders(w http.ResponseWriter, r *http.Request,
    origins []string
    }

    func isOriginAllowed(allows []string, origin string) bool {
-       for _, o := range allows {
-           if o == allOrigins {
+       origin = strings.ToLower(origin)
+
+       for _, allow := range allows {
+           if allow == allOrigins {
+               return true
+           }
+
+           allow = strings.ToLower(allow)
+           if origin == allow {
+               return true
+           }
+
+           if strings.HasSuffix(origin, o) {
+               if strings.HasSuffix(origin, "."+allow) {
+                   return true
                }
            }
        }
    }

```

**They wanted to allow
an origin and all
subdomains of the
origin...**



```

% git diff v1.4.3...v1.5.1 rest/internal/cors/handlers.go
diff --git a/rest/internal/cors/handlers.go b/rest/internal/cors/handlers.go
index e2a64b74..133b47dd 100644
--- a/rest/internal/cors/handlers.go
+++ b/rest/internal/cors/handlers.go
@@ -77,12 +77,19 @@ func checkAndSetHeaders(w http.ResponseWriter, r *http.Request,
    origins []string
    }

    func isOriginAllowed(allows []string, origin string) bool {
-       for _, o := range allows {
-           if o == allOrigins {
+       origin = strings.ToLower(origin)
+       for _, allow := range allows {
+           if allow == allOrigins {
+               return true
+           }
+
+           allow = strings.ToLower(allow)
+           if origin == allow {
+               return true
+           }
+
+           if strings.HasSuffix(origin, o) {
+               if strings.HasSuffix(origin, "."+allow) {
+                   return true
                }
            }
        }
    }

```

They actually allowed all hostnames ending with the origin



```
% git diff v1.4.3...v1.5.1 rest/internal/cors/handlers.go
diff --git a/rest/internal/cors/handlers.go b/rest/internal/cors/handlers.go
index e2a64b74..133b47dd 100644
--- a/rest/internal/cors/handlers.go
+++ b/rest/internal/cors/handlers.go
@@ -77,12 +77,19 @@ func checkAndSetHeaders(w http.ResponseWriter, r *http.Request,
origins []string
}
```

```
func isOriginAllowed(allows []string, origin string) bool {
-   for _, o := range allows {
-       if o == allOrigins {
+   origin = strings.ToLower(origin)
+   for _, allow := range allows {
+       if allow == allOrigins {
+           return true
+       }
+       allow = strings.ToLower(allow)
+       if origin == allow {
+           return true
+       }
}
```

```
-   if strings.HasSuffix(origin, o) {
+   if strings.HasSuffix(origin, "."+allow) {
+       return true
+   }
}
```

**They actually allowed all
hostnames ending with
the origin**

**pentesterlab.com ->
hackedbypentesterlab.com**



Analysing CVE (Code Review)



```
$ git show v1.4.3:rest/internal/cors/handlers.go > handlers.go-before
```



Analysing CVE (Code Review)



```
$ git show v1.4.3:rest/internal/cors/handlers.go > handlers.go-before
```

```
$ git show v1.5.1:rest/internal/cors/handlers.go > handlers.go-after
```



CVE-2008-1930





```
function wp_validate_auth_cookie($cookie = '') {
```



```
function wp_validate_auth_cookie($cookie = '') {  
    if ( empty($cookie) ) {  
        if ( empty($_COOKIE[AUTH_COOKIE]) )  
            return false;  
        $cookie = $_COOKIE[AUTH_COOKIE];  
    }  
}
```



```
function wp_validate_auth_cookie($cookie = '') {  
    if ( empty($cookie) ) {  
        if ( empty($_COOKIE[AUTH_COOKIE]) )  
            return false;  
        $cookie = $_COOKIE[AUTH_COOKIE];  
    }  
  
    list($username, $expiration, $hmac) = explode('|', $cookie);  
  
    $expired = $expiration;
```




```
function wp_validate_auth_cookie($cookie = '') {  
    if ( empty($cookie) ) {  
        if ( empty($_COOKIE[AUTH_COOKIE]) )  
            return false;  
        $cookie = $_COOKIE[AUTH_COOKIE];  
    }  
  
    list($username, $expiration, $hmac) = explode('|', $cookie);  
  
    $expired = $expiration;  
  
    // Allow a grace period for POST and AJAX requests  
    if ( defined('DOING_AJAX') ||  
        'POST' == $_SERVER['REQUEST_METHOD'] )  
        $expired += 3600;  
}
```



```
function wp_validate_auth_cookie($cookie = '') {  
    if ( empty($cookie) ) {  
        if ( empty($_COOKIE[AUTH_COOKIE]) )  
            return false;  
        $cookie = $_COOKIE[AUTH_COOKIE];  
    }  
  
    list($username, $expiration, $hmac) = explode('|', $cookie);  
  
    $expired = $expiration;  
  
    // Allow a grace period for POST and AJAX requests  
    if ( defined('DOING_AJAX') ||  
        'POST' == $_SERVER['REQUEST_METHOD'] )  
        $expired += 3600;  
  
    if ( $expired < time() )  
        return false;  
}
```



```
function wp_validate_auth_cookie($cookie = '') {  
    if ( empty($cookie) ) {  
        if ( empty($_COOKIE[AUTH_COOKIE]) )  
            return false;  
        $cookie = $_COOKIE[AUTH_COOKIE];  
    }  
  
    list($username, $expiration, $hmac) = explode('|', $cookie);  
  
    $expired = $expiration;  
  
    // Allow a grace period for POST and AJAX requests  
    if ( defined('DOING_AJAX') ||  
        'POST' == $_SERVER['REQUEST_METHOD'] )  
        $expired += 3600;  
  
    if ( $expired < time() )  
        return false;  
  
    $key = wp_hash($username . $expiration);
```




```
function wp_validate_auth_cookie($cookie = '') {
    if ( empty($cookie) ) {
        if ( empty($_COOKIE[AUTH_COOKIE]) )
            return false;
        $cookie = $_COOKIE[AUTH_COOKIE];
    }

    list($username, $expiration, $hmac) = explode('|', $cookie);

    $expired = $expiration;

    // Allow a grace period for POST and AJAX requests
    if ( defined('DOING_AJAX') ||
        'POST' == $_SERVER['REQUEST_METHOD'] )
        $expired += 3600;

    if ( $expired < time() )
        return false;

    $key = wp_hash($username . $expiration);
    $hash = hash_hmac('md5', $username . $expiration, $key);
```



```
function wp_validate_auth_cookie($cookie = '') {
    if ( empty($cookie) ) {
        if ( empty($_COOKIE[AUTH_COOKIE]) )
            return false;
        $cookie = $_COOKIE[AUTH_COOKIE];
    }

    list($username, $expiration, $hmac) = explode('|', $cookie);

    $expired = $expiration;

    // Allow a grace period for POST and AJAX requests
    if ( defined('DOING_AJAX') ||
        'POST' == $_SERVER['REQUEST_METHOD'] )
        $expired += 3600;

    if ( $expired < time() )
        return false;

    $key = wp_hash($username . $expiration);
    $hash = hash_hmac('md5', $username . $expiration, $key);

    if ( $hmac != $hash )
        return false;
}
```



```
function wp_validate_auth_cookie($cookie = '') {
    if ( empty($cookie) ) {
        if ( empty($_COOKIE[AUTH_COOKIE]) )
            return false;
        $cookie = $_COOKIE[AUTH_COOKIE];
    }

    list($username, $expiration, $hmac) = explode('|', $cookie);

    $expired = $expiration;

    // Allow a grace period for POST and AJAX requests
    if ( defined('DOING_AJAX') ||
        'POST' == $_SERVER['REQUEST_METHOD'] )
        $expired += 3600;

    if ( $expired < time() )
        return false;

    $key = wp_hash($username . $expiration);
    $hash = hash_hmac('md5', $username . $expiration, $key);

    if ( $hmac != $hash )
        return false;

    $user = get_userdata_by_login($username);
    if ( ! $user )
        return false;

    return $user->ID;
}
```



admin:1353464343:16849b89783b5918a41bbd29a3c4bbf6

admin
1353464343
16849b89783b5918a41bbd29a3c4bbf6

hmac(admin1353464343)

```
function wp_validate_auth_cookie($cookie = '') {
    if ( empty($cookie) ) {
        if ( empty($_COOKIE[AUTH_COOKIE]) )
            return false;
        $cookie = $_COOKIE[AUTH_COOKIE];
    }

    list($username, $expiration, $hmac) = explode('|', $cookie);

    $expired = $expiration;

    // Allow a grace period for POST and AJAX requests
    if ( defined('DOING_AJAX') ||
        'POST' == $_SERVER['REQUEST_METHOD'] )
        $expired += 3600;

    if ( $expired < time() )
        return false;

    $key = wp_hash($username . $expiration);
    $hash = hash_hmac('md5', $username . $expiration, $key);

    if ( $hmac != $hash )
        return false;

    $user = get_userdata_bylogin($username);
    if ( ! $user )
        return false;

    return $user->ID;
}
```

admin1:1353464343:1ba7d82099dd6119781b54ecf8b79259

admin1
1353464343
1ba7d82099dd6119781b54ecf8b79259

hmac(admin11353464343)

```
function wp_validate_auth_cookie($cookie = '') {
    if ( empty($cookie) ) {
        if ( empty($_COOKIE[AUTH_COOKIE]) )
            return false;
        $cookie = $_COOKIE[AUTH_COOKIE];
    }

    list($username, $expiration, $hmac) = explode('|', $cookie);

    $expired = $expiration;

    // Allow a grace period for POST and AJAX requests
    if ( defined('DOING_AJAX') ||
        'POST' == $_SERVER['REQUEST_METHOD'] )
        $expired += 3600;

    if ( $expired < time() )
        return false;

    $key = wp_hash($username . $expiration);
    $hash = hash_hmac('md5', $username . $expiration, $key);

    if ( $hmac != $hash )
        return false;

    $user = get_userdata_bylogin($username);
    if ( ! $user )
        return false;

    return $user->ID;
}
```

admin**1**:1353464343:1ba7d82099dd6119781b54ecf8b79259
admin:**11353464343**:1ba7d82099dd6119781b54ecf8b79259

admin
11353464343
1ba7d82099dd6119781b54ecf8b79259

hmac(admin11353464343)

hmac(admin**11353464343**)

```
function wp_validate_auth_cookie($cookie = '') {
    if ( empty($cookie) ) {
        if ( empty($_COOKIE[AUTH_COOKIE]) )
            return false;
        $cookie = $_COOKIE[AUTH_COOKIE];
    }

    list($username, $expiration, $hmac) = explode('|', $cookie);

    $expired = $expiration;

    // Allow a grace period for POST and AJAX requests
    if ( defined('DOING_AJAX') ||
        'POST' == $_SERVER['REQUEST_METHOD'] )
        $expired += 3600;

    if ( $expired < time() )
        return false;

    $key = wp_hash($username . $expiration);
    $hash = hash_hmac('md5', $username . $expiration, $key);

    if ( $hmac != $hash )
        return false;

    $user = get_userdata_bylogin($username);
    if ( ! $user )
        return false;

    return $user->ID;
}
```


The Fix



```
- $key = wp_hash($username . $expiration);  
- $hash = hash_hmac('md5', $username . $expiration, $key);  
+ $key = wp_hash($username . '|' . $expiration);  
+ $hash = hash_hmac('md5', $username . '|' . $expiration, $key);
```

Lesson learned:
**Always include a delimiter between values
when signing data.**



CONCLUSION 🧐



Assumptions!

Developers, yours, ...







“All important targets require substantial initial investments before discovering and consistently discovering vulnerabilities.”

- Silvio Cesare



**If you create software, you
may have as many CVEs as you want**



**And don't forget to
add them all to your CV!**



imgflip.com



Keeping in touch



<https://pentesterlab.com/>



<https://pentesterlab.com/live-training/>



louis@pentesterlab.com



[@pentesterlab.com](https://twitter.com/pentesterlab) and [@snyff.pentesterlab.com](https://twitter.com/snyff.pentesterlab)



<https://www.linkedin.com/company/pentesterlab/>



<https://www.linkedin.com/in/snyff/>



[@PentesterLab](https://twitter.com/PentesterLab) and [@snyff](https://twitter.com/snyff)



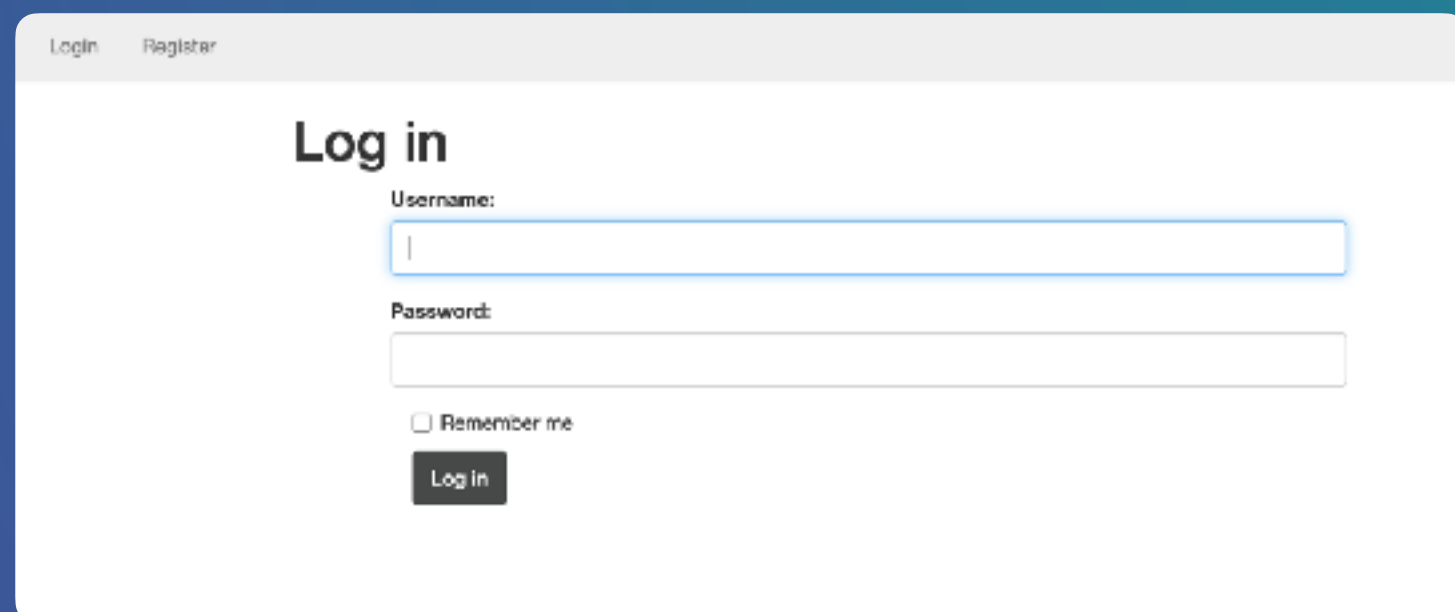
Conclusion

- Practice makes perfect
- There are still ****PLENTY**** of bugs to be found
- Keep notes!
- Now it's time to review some code!



Hands-On 🤝

- One application written in both Golang and PHP:
 - PHP: <https://github.com/PentesterLab/codereview-php>
 - Golang: <https://github.com/PentesterLab/codereview-golang>
- A lot of vulnerabilities...



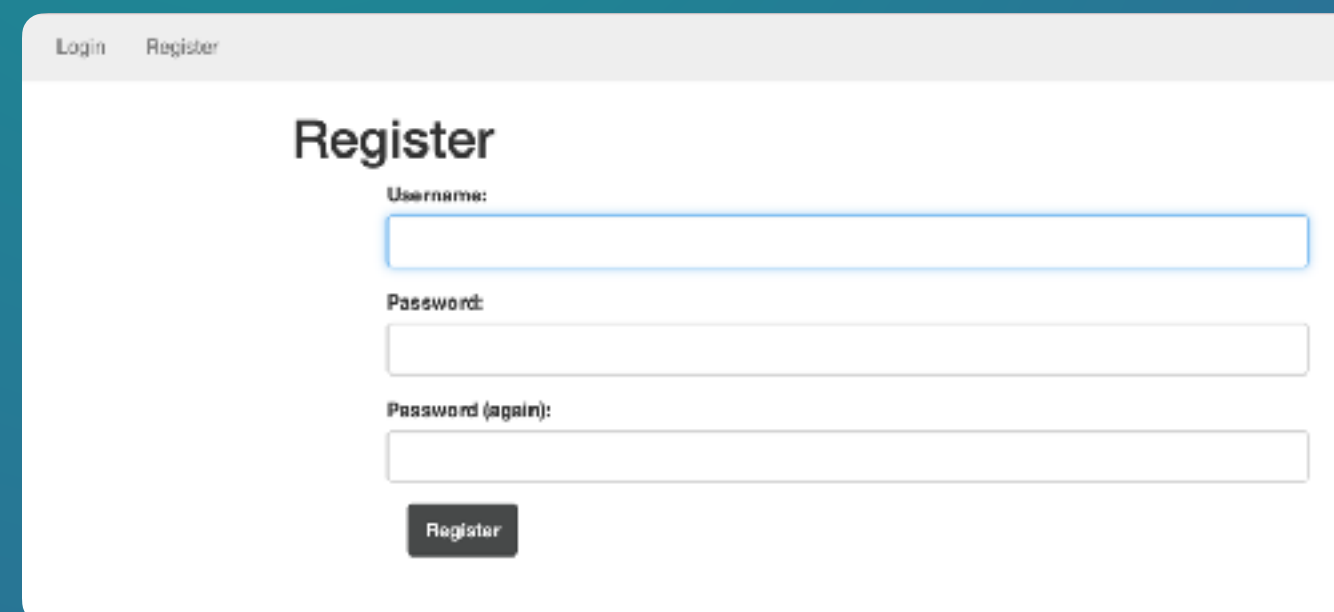
Log in Register

Log in

Username:

Password:

☐ Remember me



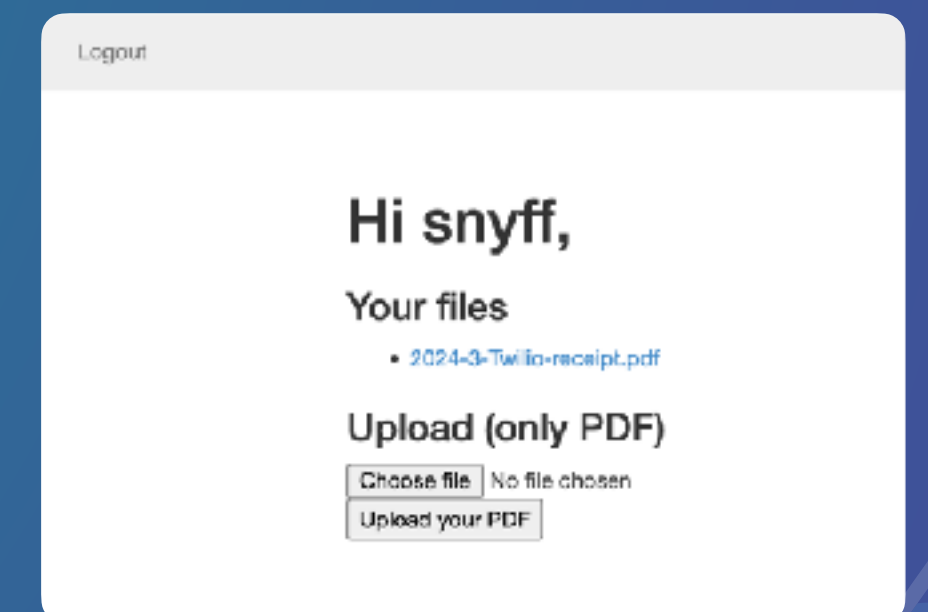
Login Register

Register

Username:

Password:

Password (again):



Logout

Hi snyff,

Your files

- 2024-3-Twilio-receipt.pdf

Upload (only PDF)

No file chosen



Hands-On 🤝: Code Review!

Login Register

Register

Username:

Password:

Password (again):

Register

Login Register

Log in

Username:

Password:

☐ Remember me

Log in

Logout

Hi snyff,

Your files

- [2024-3-Twilio-receipt.pdf](#)

Upload (only PDF)

Choose file No file chosen

Upload your PDF

